Week 6 Solution

NPTEL PSOSM Course (Privacy and Security in Online Social Media)

Question 1

Consider the following scenario on a social network:

A is friends with B and C.

B is friends with A and D.

C is friends with A and D.

D is friends with B and C.

Calculate the betweenness centrality for A, B, C and D. Identify the individual having the highest betweenness centrality.

(a) A

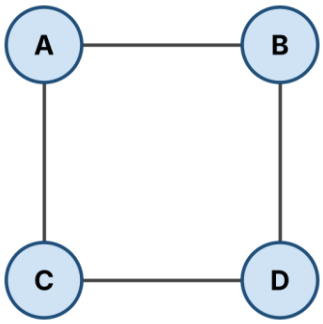
(b) B

(c) C

(d) All of them have the same betweenness centrality ✓

Solution

Step 1: Draw the graph. Friendship is mutual, so the network is undirected with edges A-B, A-C, B-D and C-D. These four edges form a single cycle A-B-D-C-A (a 4-cycle, or square).



Step 2: Formula. Betweenness centrality of a node v is

$$C_B(v) = \Sigma \text{ over pairs } (s, t),\ s \neq v \neq t, \text{ of } \sigma_{st}(v) / \sigma_{st}$$

where σ_st is the number of shortest paths between s and t, and σ_st(v) is the number of those shortest paths that pass through v. For an undirected graph each unordered pair is counted once.

Step 3: List all shortest paths. There are 4C2 = 6 pairs of nodes.

Pair	Shortest paths	Count σ	via A	via B	via C	via D
A-B	A-B (direct)	1	-	-	0	0
A-C	A-C (direct)	1	-	0	-	0
A-D	A-B-D, A-C-D	2	-	1/2	1/2	-
B-C	B-A-C, B-D-C	2	1/2	-	-	1/2
B-D	B-D (direct)	1	0	-	0	-
C-D	C-D (direct)	1	0	0	-	-

Total			0.5	0.5	0.5	0.5

Step 4: Interpret. A, B, C and D each lie on exactly one of the two equally short paths of one pair, so each gets 1/2 = 0.5. Normalised by (n-1)(n-2)/2 = 3, each value is 0.5/3 = 1/6. Since the 4-cycle is perfectly symmetric (every node looks identical), no node can be more 'in between' than another.

Answer: All of them have the same betweenness centrality (each = 0.5, normalised 1/6).

Question 2

What does indegree centrality measure in the social media and network analysis context?

(a) The level of activity a user has on the platform.

(b) The number of outgoing connections a user has with other users.

(c) A user's popularity based on their followers or incoming connections. ✓

(d) The diversity of content shared by a user.

Solution

In a directed network (such as Twitter or Instagram, where following need not be mutual), every node has two kinds of degree:

- **Indegree:** number of edges pointing INTO the node, that is, the number of followers.
- **Outdegree:** number of edges going OUT of the node, that is, the number of accounts the user follows.

A user with many incoming edges is one whom many others chose to follow, so indegree centrality is a direct measure of popularity or prestige. For example, a celebrity with 10 million followers but following only 200 has very high indegree and low outdegree.

Why the others are wrong: option (b) describes outdegree; options (a) and (d) are about activity and content, which are not structural graph measures.

Answer: A user's popularity based on their followers or incoming connections.

Question 3

What does betweenness centrality depict about the node?

(a) One who is quickly approachable ✓

(b) Who is close to everyone

(c) One who is most influential

(d) All of the above

Solution

Betweenness centrality counts how often a node lies on the shortest paths between other pairs of nodes. A node with high betweenness sits on the 'highways' of the network: information travelling between many pairs of people passes through it, so it is the person through whom others can be reached quickly. In this course's terminology, that node is described as quickly approachable (a bridge or broker).

Why the other options are not accepted:

- **Who is close to everyone:** this is closeness centrality, based on the average shortest-path distance from a node to all others.
- **Most influential:** this is usually associated with eigenvector centrality or PageRank, where a node is important if it is connected to other important nodes.

Answer: One who is quickly approachable (the accepted answer).

Question 4

What is link farming?

(a) Agricultural practice that involves cultivating crops in a vertical arrangement.

(b) Rapidly growing links on a website through unethical or manipulative means. ✓

(c) A strategy for developing social connections by attending networking events.

(d) All of the above

Solution

Search engines such as Google (PageRank) rank a page higher when many other pages link to it. A link farm is a group of websites (or accounts) that all link to one another, or to a target, purely to inflate this link count artificially. The links carry no genuine endorsement.

On social media the same idea appears as follower farming: spammers follow large numbers of users hoping they follow back, so that the spammer's follower count and apparent influence grow quickly. Search engines treat link farming as a black-hat SEO technique and penalise sites that use it.

Option (a) is a distractor (that is vertical farming), and option (c) describes legitimate networking.

Answer: Rapidly growing links on a website through unethical or manipulative means.

Question 5

Spamming the index of search engines is called (multiple correct)

(a) Spamdexing ✓

(b) Spamexing ✓

(c) Spamming

(d) None of the above

Solution

The word is a blend of 'spam' and 'indexing'. Spamdexing (also written spamexing, and also called search engine spam or search engine poisoning) is the deliberate manipulation of a search engine's index so that a page ranks higher than it deserves. Common techniques:

- Keyword stuffing: repeating popular keywords many times, often in hidden text.
- Hidden or invisible text: text in the same colour as the background.
- Link farms and link spam (see Question 4).
- Cloaking: showing one page to the search engine crawler and another to human visitors.

'Spamming' alone is too general (it covers email spam, comment spam and so on), so it is not the specific term.

Answer: Spamdexing and Spamexing (both accepted).

Question 6

Which of the following is the best example of a clickbait headline?

(a) Annual Financial Report 2026

(b) Meeting Agenda for Today

(c) You Won't Believe What Happened Next! Click Here to Claim Your Prize! ✓

(d) Python Programming Notes

Solution

Clickbait is content whose main purpose is to attract clicks, usually by exploiting curiosity or excitement rather than accurately describing the content. Typical signs:

- Curiosity gap: hides the key information ('You won't believe what happened next').
- Sensational or exaggerated language and exclamation marks.
- Direct calls to action and lures such as 'Click here' or 'Claim your prize'.

Option (c) has all three signs. Options (a), (b) and (d) are plain, descriptive titles that tell the reader exactly what they will get.

Answer: You Won't Believe What Happened Next! Click Here to Claim Your Prize!

Question 7

Which of the following is considered as 'social etiquette' on online social media such as Twitter?

(a) Following back ✓

(b) Comment on the post of an influential person

(c) Giving likes to the post of an influential person

(d) Saying 'Thank you' to any appreciation by an influential person

Solution

On Twitter many users consider it polite to follow back someone who follows them. This unwritten norm is called social etiquette and is driven by reciprocity (see Question 8).

This etiquette is exactly what link farmers exploit: a spammer follows thousands of users, many of whom follow back out of courtesy, and the spammer gains a large follower base quickly. Users who routinely follow back are called social capitalists in the link farming literature.

Options (b), (c) and (d) are optional forms of engagement, not the established norm that the network relies on.

Answer: Following back.

Question 8

A psychological principle where individuals feel compelled to return favours and interactions they receive is called

(a) Spamming

(b) Reciprocity ✓

(c) Link farming

(d) All of the above

Solution

Reciprocity (the norm of reciprocity, one of Cialdini's principles of persuasion) is the human tendency to return what we receive: a favour for a favour, a follow for a follow, a like for a like.

Connection to the other options: reciprocity is the psychology; following back (Question 7) is the etiquette it creates; link farming and spamming are the manipulative techniques that exploit it. So spamming and link farming are not the principle itself, which rules out (a), (c) and (d).

Answer: Reciprocity.

Question 9

Which of the following is/are graph centrality measures? (multiple correct)

(a) Eigen centrality ✓

(b) Node degree ✓

(c) Betweenness centrality ✓

(d) Closeness centrality ✓

Solution

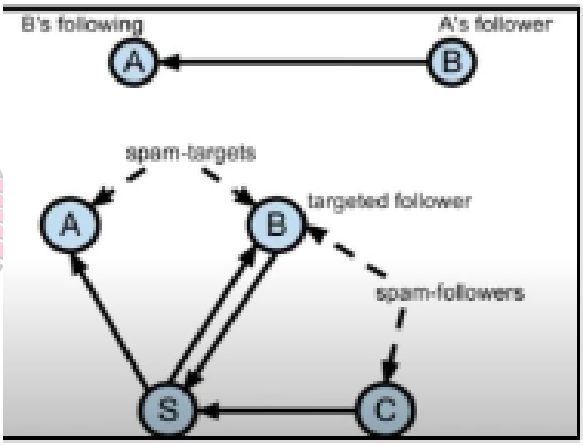
A centrality measure assigns each node a score that captures how important or central it is. All four options are standard centrality measures:

Measure	Idea	Formula (n nodes)
Degree centrality	Number of direct connections	$C_D(v) = deg(v) / (n-1)$
Closeness centrality	How near a node is to all others	$C_C(v) = (n-1) / \Sigma d(v,u)$
Betweenness centrality	How often a node lies on shortest paths	$C_B(v) = \Sigma \sigma_{st}(v) / \sigma_{st}$
Eigenvector centrality	Important if connected to important nodes	$A x = \lambda x$ (leading eigenvector)

Answer: Eigen centrality, Node degree, Betweenness centrality and Closeness centrality (all four).

Question 10

Which of the following are spam targets? (multiple correct)



(a) A ✓

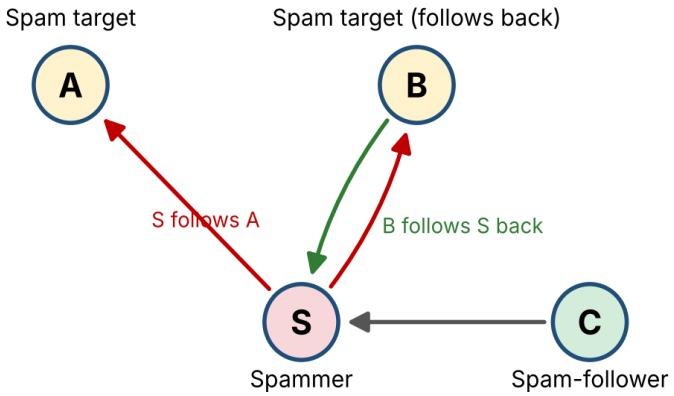
(b) B ✓

(c) C

(d) S

Solution

The figure models link farming on Twitter. Reading the roles from the arrows:



- **S (spammer):** the account trying to gain followers. It sends follow requests outward.
- **A and B (spam targets):** the dashed arrows labelled 'spam-targets' point to A and B, and S has follow edges going to both. These are the legitimate users the spammer follows in the hope of being followed back. B did follow back (the edge B to S), so B is also labelled the 'targeted follower'.
- **C (spam-follower):** C follows S, but S did not target C; C is a follower the spammer gained (labelled 'spam-followers').

So the targets of the spammer's follow requests are A and B. C is a follower, and S is the spammer itself, so neither of them is a target.

Answer: A and B.

Answer Key

Q.No.	Correct answer
1	All of them have the same betweenness centrality
2	A user's popularity based on followers or incoming connections
3	One who is quickly approachable
4	Rapidly growing links through unethical or manipulative means
5	Spamdexing, Spamexing
6	You Won't Believe What Happened Next! Click Here to Claim Your Prize!
7	Following back
8	Reciprocity
9	Eigen, Degree, Betweenness, Closeness (all four)
10	A, B